

Of Synthetic Realms, Discontinuities, and Shift-Shares

Lecture 3: Synthetic Control, RDD, and Shift-Share

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Synthetic Control

Regression Discontinuity Design

Shift-Share IV

Lab Preview

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The single-treated-unit problem

When DiD breaks down

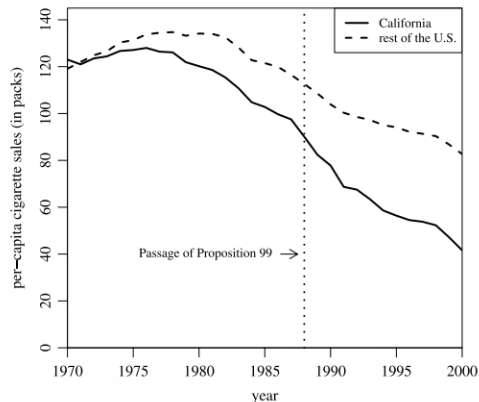
- DiD needs many treated units for inference
- Policy often applies to *one* state, country, or city
- Who is the right control group?
- Researcher discretion $\Rightarrow$ cherry-picking risk

The question (Abadie et al., 2010)

California passes Proposition 99 in 1988: a large tobacco tax. *What would cigarette consumption have been without it?*

The challenge

No single state is a good control. Using all states or a researcher-chosen subset gives different answers.



Why not "rest of US"? The naive control (dashed) trends *upward* while CA declines pre-1988 — parallel trends fails.

Synthetic Control: building the counterfactual

Setup (Abadie et al., 2010)

- $J + 1$ units, T periods; unit 1 treated at T_0
- *Donor pool*: units $2, \dots, J + 1$ never treated
- *Synthetic control*: $\hat{Y}_{1t}^{(0)} = \sum_{j=2}^{J+1} w_j^* Y_{jt}$
- Weights: $w_j^* \geq 0, \sum w_j^* = 1$

How weights are chosen

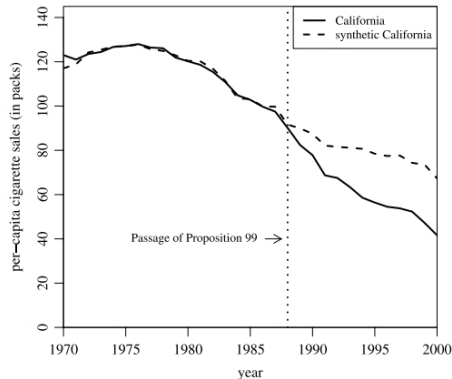
Minimise pre-treatment prediction error:

$$W^* = \arg \min_W \|X_1 - X_0 W\|_V$$

X_1, X_0 : outcome lags + covariates; V : importance matrix.

Key property

$w_j^* \geq 0, \sum w_j^* = 1 \Rightarrow$ interpolation only; no extrapolation outside the convex hull of donors.



CA vs synthetic CA, 1970–2000: tight pre-fit, sharp post-1988 divergence = treatment effect.

Sparse weights: CT (0.69), CO (0.16), NV (0.09), MT (0.05), UT (0.01)

Pre-fit RMSPE: synth ≈ 1.6 vs rest-of-US ≈ 8.0 packs.

Inference: the placebo (permutation) test (**Abadie et al., 2010**)

Why standard errors fail

One treated unit $\Rightarrow$ no asymptotic distribution.
Standard bootstrapping and DiD SEs both require many units.

Placebo logic (**Abadie et al., 2010**): apply synth to **each donor state** as if treated in 1988, then rank states by the post/pre RMSPE ratio:

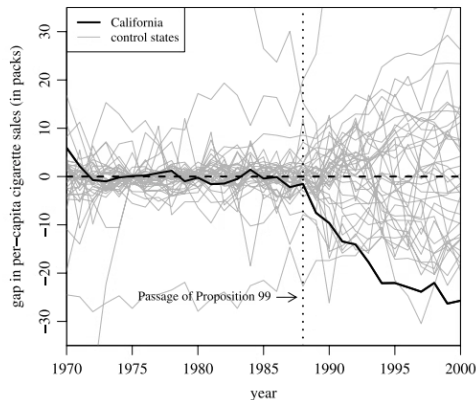
$$r_j = \frac{\text{RMSPE}_{\text{post}}}{\text{RMSPE}_{\text{pre}}}$$

Dividing by pre-RMSPE controls for poor pre-fit.

p-value = share of states with $r_j \geq r_{\text{CA}}$.

Result

CA ranks **1st out of 39 states**: $p = 1/39 \approx 0.026$.



Gray: gap (actual — synthetic) for each placebo state. Black: California. Pre-1988 all gaps cluster near zero; post-1988 CA drops to -26 packs — far below every placebo. That is the visual p -value.

When synth works well

- Aggregate-level treatment (state, country, city)
- Long pre-treatment period ($T_0 \geq 10$ periods)
- Good pre-treatment fit available from donor pool
- Single or few treated units
- Policy well-defined with known timing

When synth fails

- No good match in donor pool (large pre-RMSPE)
- Donor units also partially treated or anticipate treatment
- Short pre-period $\Rightarrow$ over-fit
- Spillovers from treated to donors

Synth vs. DiD — the key trade-off

- DiD requires **parallel trends**
- Synth replaces this with **convex hull inclusion**: the treated unit's pre-treatment path must lie in the convex hull of the donor pool $\Rightarrow$ *less stringent*
- If the treated unit is an outlier that no weighted combination of donors can reproduce, pre-fit will be poor and the estimate invalid

Bridging Synth and DiD (Arkhangelsky et al., 2021)

Both methods share a flaw:

- DiD: parallel trends may not hold
- Synth: treated unit must lie in the convex hull of donors

SDID solves both by combining

- *Unit weights* $\hat{\omega}_j$ — like synthetic control, re-weight donors to match treated pre-trends
- *Time weights* $\hat{\lambda}_t$ — up-weight pre-treatment periods that look most like the treatment period

Estimator

$$\hat{\tau}^{\text{SDID}} = \arg \min_{\tau, \alpha, \beta} \sum_{i,t} \hat{\omega}_i \hat{\lambda}_t (Y_{it} - \alpha_i - \beta_t - \tau D_{it})^2$$

Key properties

- Consistent even if trends are *not* parallel
- Works with multiple treated units (unlike basic synth)
- Nests DiD ($\hat{\omega}_j = 1/N_{\text{ctrl}}$) and SC ($\hat{\lambda}_t = 1/T_{\text{pre}}$)
- Bootstrap/placebo inference

When to prefer SDID over Synth

Many treated units; possible pre-trend divergence; want standard errors with a formula (not just permutation).

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The design

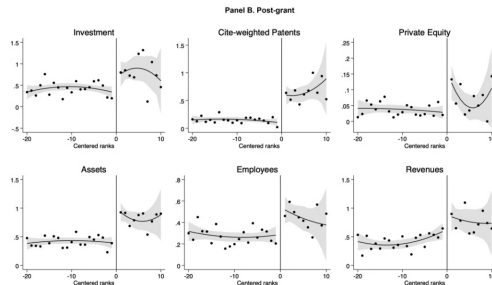
- Running variable X_i (score, vote margin, ...)
- Threshold c : $D_i = \mathbf{1}[X_i \geq c]$
- Units near c are locally comparable
- Assignment “as good as random” near cutoff

Design

- **Programme:** EU SME Instrument (Horizon 2020) — first European R&D grant targeting small innovative firms; grants up to €2.5m
- **Running variable:** expert panel score rank (centered at funding threshold)
- **Treatment:** grant awarded ($\text{rank} \geq 0$)
- **Sample:** firms from >40 countries, virtually all sectors

Key results

- **+15%–31%** cite-weighted patents (intensive & extensive margins)
- **>100%** increase in probability of receiving follow-on private equity
- Effects larger for younger, smaller firms and financially constrained sectors/regions



Panel B (post-grant): RD plots at the funding threshold (centered rank = 0). Clear jumps for investment, cite-weighted patents, and private equity. Threshold assignment identifies the causal effect.

Identification: the continuity assumption

Formal assumption

The potential outcome functions

$$E[Y_i^{(0)} | X_i = x] \text{ and } E[Y_i^{(1)} | X_i = x]$$

are **continuous** in x at c .

What this buys

$$\hat{\tau}_{\text{RDD}} = \lim_{x \downarrow c} E[Y | X = x] - \lim_{x \uparrow c} E[Y | X = x]$$

Identifies the **Local Average Treatment Effect** at c .

What it doesn't

- $\text{LATE} \neq \text{ATE}$ (effect may vary away from threshold)
- Cannot extrapolate away from c
- Says nothing about why the discontinuity holds

Key violation: manipulation

If units sort above/below c , X_i is not continuously distributed there. Examples: students repeating exams, parties inflating vote counts, firms manipulating ratios.

Local vs. global

RDD is *local*: only observations near c identify τ . This is a feature (credibility) and a limitation (external validity).

RD plots: key decisions (Cattaneo et al., 2019)

Visualise the jump at the cutoff. The plot uses *binned sample means* (dots) plus a *global polynomial smooth* (line) separately on each side.

Decision 1 — Bin type

- **Evenly-spaced (ES):** bins of equal width. Works well when the score is roughly uniform.
- **Quantile-spaced (QS):** bins with equal *number of observations*. Better when the score is skewed or uneven — avoids bins with very few points.

Decision 2 — Number of bins

- **IMSE-optimal:** smooth plot showing overall regression shape.
- **MV (mimicking variance):** more bins → to see scatter around the trend near the cutoff.

Decision 3 — Global polynomial order

The smooth line uses a *global* polynomial (default order 4) to illustrate the overall shape → for *visualisation only*, not for the formal estimate.

Recommended default: QS + MV

Quantile-spaced bins + MV number of bins shows the local variability and handles uneven score distributions. → can switch to IMSE to assess the global shape of the regression function.

What to look for: a clear jump in the binned means at c . If dots on the two sides overlap near c → effect small or absent.

The local linear estimator ($p = 1$)

Within bandwidth h on each side of c , fit:

$$Y_i = \alpha + \tau D_i + \beta(X_i - c) + \gamma D_i(X_i - c) + \varepsilon_i$$

The intercept jump $\hat{\tau}$ is the RD estimate.

Decision 1 — Polynomial order p

- **Local linear** ($p = 1$): *recommended default*. Robust to boundary problems; no overfitting.
- **Local quadratic** ($p = 2$): used for bias-correction only (no estimate).
- **Global polynomials**: never use $\rightarrow$ poor at boundaries; misleading results.

Decision 2 — Kernel function $K(\cdot)$

- **Triangular**: more weight to observations closer to c . *Recommended*
- **Uniform**: equal weights $\rightarrow$ only with very few observations near the cutoff.

Decision 3 — Bandwidth h

- **MSE-optimal** (Calonico et al., 2014): balances bias and variance. *Use for point estimation*.
- **CER-optimal**: smaller bandwidth that minimises coverage error of confidence intervals. *Use for inference* when coverage accuracy matters most.
- **Two bandwidths** (h_- , h_+): allowed when curvature or density differs across sides.

Decision 4 — Inference: bias-correction and robust CIs

- point estimator has *non-negligible bias* $\Rightarrow$ CIs are incorrect $\rightarrow$ Calonico et al. (2014): estimate the bias using a local polynomial of order $p + 1$ and subtract it

Decision 5 — Predetermined covariates

- Including covariates only improves *precision* by reducing residual variance.
- Only use **pre-determined** covariates
- Check: covariates should show *no jump* at c (balance test); if they do, the RD assumption fails.

Decision 6 — Clustering

- Cluster if observations within a group share common unobservables (e.g., same school, district, year).

Rule of thumb

$p = 1$ + triangular kernel + MSE bandwidth + robust CI. Always report bandwidth sensitivity.

Validity: what you must always check

1. Density test

Is the density of X_i continuous at c ?

A jump $\Rightarrow$ units are sorting; running variable is manipulated. *E.g. firms inflating R&D spend just above the eligibility threshold to qualify for a grant.*

2. Covariate balance

Run RDD on pre-determined covariates.

Should find **no jump** at c .

3. Placebo outcomes (Cattaneo et al., 2019)

Outcomes measured *after* treatment but causally unrelated to it — should find **no effect**.

Clean water: car accident mortality is a valid placebo; water-borne illness mortality is not.

4. Placebo cutoffs

Test at arbitrary thresholds $c' \neq c$ within the support.

Should find **no significant effect**

5. Bandwidth sensitivity

Plot $\hat{\tau}(h)$ for a range of bandwidths.

Stable $\Rightarrow$ reassuring

6. Donut RDD

Exclude obs. within a small radius δ of c .

Tests whether the effect is driven by near-threshold manipulation. Effect should persist with $\delta > 0$.

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The shift-share instrument

$$z_{\ell} = \sum_{n=1}^N \underbrace{s_{\ell n}}_{\text{shares}} \cdot \underbrace{g_n}_{\text{shocks}}$$

- $s_{\ell n}$: lagged employment share of industry n in region ℓ
- g_n : national (leave-one-out) employment growth in industry n
- z_{ℓ} : predicted local employment growth from national trends

Idea (Bartik, 1991)

Two regions with different industry mixes receive *different shocks* even if the national trend is the same — generating exogenous variation in the endogenous treatment.

Notation

Model: $y_{\ell} = \beta x_{\ell} + \gamma' w_{\ell} + \varepsilon_{\ell}$

Use z_{ℓ} as IV for the endogenous x_{ℓ} .

Validity: $E[z_{\ell} \varepsilon_{\ell}] = 0$

Three canonical applications

Paper	Shares $s_{\ell n}$	Shocks g_n	Outcome y_{ℓ}	Data level
Bartik (1991)	Lagged industry empl. shares	Natl. empl. growth by industry	Local employment growth	MSAs
ADH (2013)	Mfg. empl. shares (10-yr lag)	China exports to 8 non-US countries	Mfg. employment & wages	Commuting zones
Card (2009)	Migrant enclaves (past shares)	Natl. immigration by origin	Native wages by education	Cities/MSAs

Two ways to justify SSIV

Strategy 1: Shock exogeneity (BHJ)

Borusyak et al. (2022): exogeneity lives in **shocks** g_n .

Assumption: g_n quasi-randomly assigned across industries, conditional on controls.

Implication: shares $s_{\ell n}$ are just *exposure weights* — need not be exogenous.

Test: pre-trend balance at the industry level.

Inference: cluster at the industry (shock) level.

Strategy 2: Share exogeneity (GPSS)

Goldsmith-Pinkham et al. (2020): exogeneity lives in **shares** $s_{\ell n}$.

Assumption: $s_{\ell n}$ pre-determined and uncorrelated with ε_{ℓ} .

Implication: SSIV = weighted avg of industry IVs with Rotemberg weights α_n :

$$\hat{\beta} = \sum_n \alpha_n \hat{\beta}_n$$

Test: shares uncorrelated with pre-trends? Rotemberg weights non-negative?

Two views — one instrument

Which view you take determines what you must defend. BHJ: many quasi-random shocks. GPSS: clean historical shares. Both require the exclusion restriction.

Shock exogeneity (BHJ) — three checks

1. Leave-one-out shocks

Exclude focal region from its own shock to avoid mechanical correlation:

$$g_n^{(-\ell)} = \frac{\Delta E_n - \Delta E_{\ell n}}{E_n - E_{\ell n}}$$

2. Cluster at industry level

SEs must account for shock-level correlation (Adaõ et al., 2019). Industries induce correlated residuals across regions with similar shares — cluster at the shock, not the region.

3. Industry-level balance test

Regress g_n on pre-determined industry characteristics. Significant predictors $\Rightarrow$ shocks not quasi-randomly assigned.

Share exogeneity (GPSS) — three checks

1. Share pre-trends

For the highest- α_n industries, test whether $s_{\ell n}$ predicts the outcome *before* treatment. Significant pre-trends $\Rightarrow$ shares not pre-determined.

2. Rotemberg weights

Compute α_n (each industry's contribution to $\hat{\beta}$).

Check: (a) all $\alpha_n \geq 0$ — negative weights signal fragility; (b) no single industry dominates — concentration means the result rests on one sector's exogeneity (Drop top 3 industries; re-estimate. Stable $\rightarrow$ robust. Flips $\rightarrow$ fragile).

3. Incomplete shares

If $\sum_n s_{\ell n} < 1$ (e.g. only manufacturing covered): normalise to sum to 1, or control for coverage share.

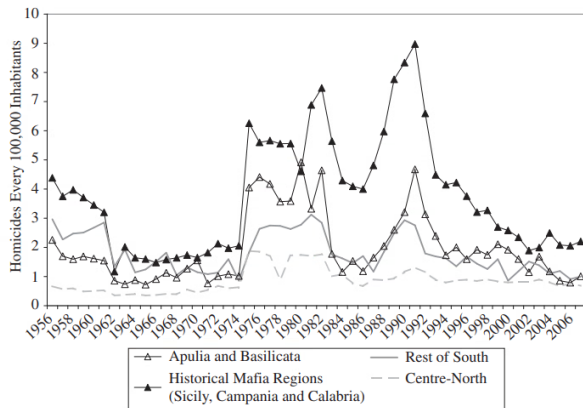
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Exercise 1: mafia expands into Apulia & Basilicata (Pinotti, 2015)



Research question: causal economic cost of organised crime?

Two exogenous drivers:

1. **Tobacco route shift:** Tangier port closure (1960) ⇒ depots move to E. Europe ⇒ syndicates switch from *Tyrrhenian* (Morocco–Naples) to *Adriatic route* (Albania–Apulia). Cigarette smuggling was the dominant criminal business of the 1970s.
2. **Confino:** convicted mafia members forcibly relocated by government; Basilicata had the most in Italy (1961–72).

Why this matters for synth

Onset **dated** (homicides spike from 1975), **abrupt**, and **exogenous** — driven by port policy and geography, not regional choice.

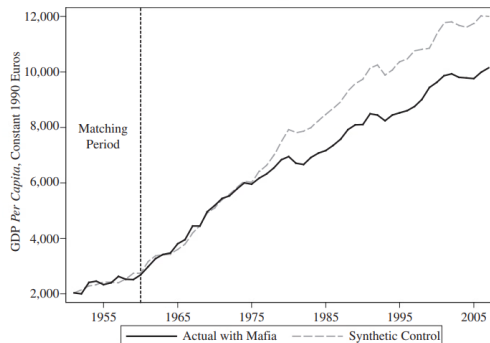
Application: the economic cost of organised crime (Pinotti, 2015)

Setup

- **Treated:** Apulia & Basilicata — free of mafia until rapid expansion in the early 1970s
- **Donor pool:** other Italian regions without mafia
- **Outcome:** GDP per capita (constant 1990 euros)
- **Pre-period:** post-war to early 1960s (matching period)

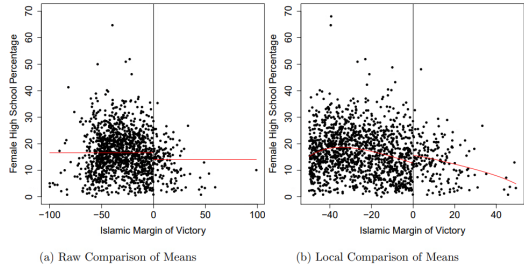
Why synth fits & key result

- Dated treatment onset, long clean pre-period
- $N_{\text{treated}} = 2$ — too small for DiD inference
- Donor units unaffected by the same shock
- Mafia $\Rightarrow$ -16% **GDP per capita**. Mechanism: private investment crowded out by less productive public spending



Exercise 2: Turkish municipal elections and female higher education completion rate (Meyersson, 2014)

- 1994 Turkish municipal elections, $n \approx 2,629$
- Running var: Islamic party vote *margin*.
- Treatment: Islamic mayor elected ($X_i \geq 0$).
- Outcome: female HS completion rate (2000).
- **Finding:** +4.4 pp for women in Islamic-won municipalities.



(a) Full bandwidth: no jump visible. (b) Local bandwidth: clear discontinuity at $c = 0$.

Why does an Islamic mayor increase female HS completion? — main channels

- (1) **Supply:** more schools built $\Rightarrow$ shorter travel distance, a binding constraint for girls in conservative households
- (2) **Legitimacy:** conservative families more willing to allow daughters to attend under a religiously aligned government
- (3) **Safety:** reduced social barriers (harassment, stigma) for girls commuting.
- (4) Effect is largest in poorer, more conservative areas $\rightarrow$ consistent with supply + social-norms channels.

Exercise 3: the China shock and US manufacturing employment (Autor et al., 2013)

- 722 US commuting zones, two decades (1990s, 2000s)
- **Endogenous:** Δ Chinese imports/worker (x_ℓ)
- **Instrument:** $z_\ell = \sum_n s_{\ell n}^{\text{lag}} \cdot g_n^{\text{other}}$ — lagged CZ industry shares $\times$ China exports to 8 other countries
- **Outcome:** Δ manufacturing employment share
- **Finding:** \$1,000 more imports/worker $\Rightarrow$ -0.6 pp mfg employment

Why the IV works

- CZs with different industry mixes are hit differently by the same global shock.
- China exports to *other* countries are exogenous to US local conditions.
- 2SLS $\gg$ OLS $\Rightarrow$ OLS is biased toward zero by two forces:
 - $\rightarrow$ **Measurement error:** industry misclassification in US customs data adds noise to $x_\ell \Rightarrow$ attenuation
 - $\rightarrow$ **Simultaneity:** structurally declining regions attract cheap imports $\Rightarrow$ reverse causality also shrinks the OLS

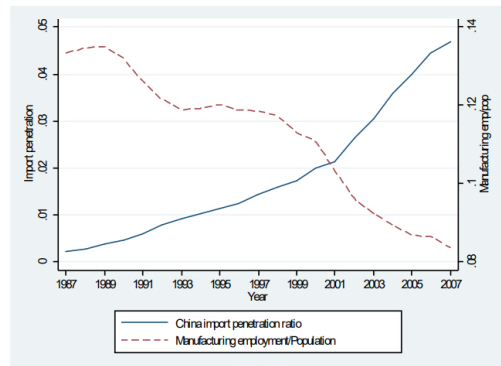


Figure 1.
Import Penetration Ratio for U.S. Imports from China (left scale), and Share of U.S. Working-Age Population Employed in Manufacturing (right scale).

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